

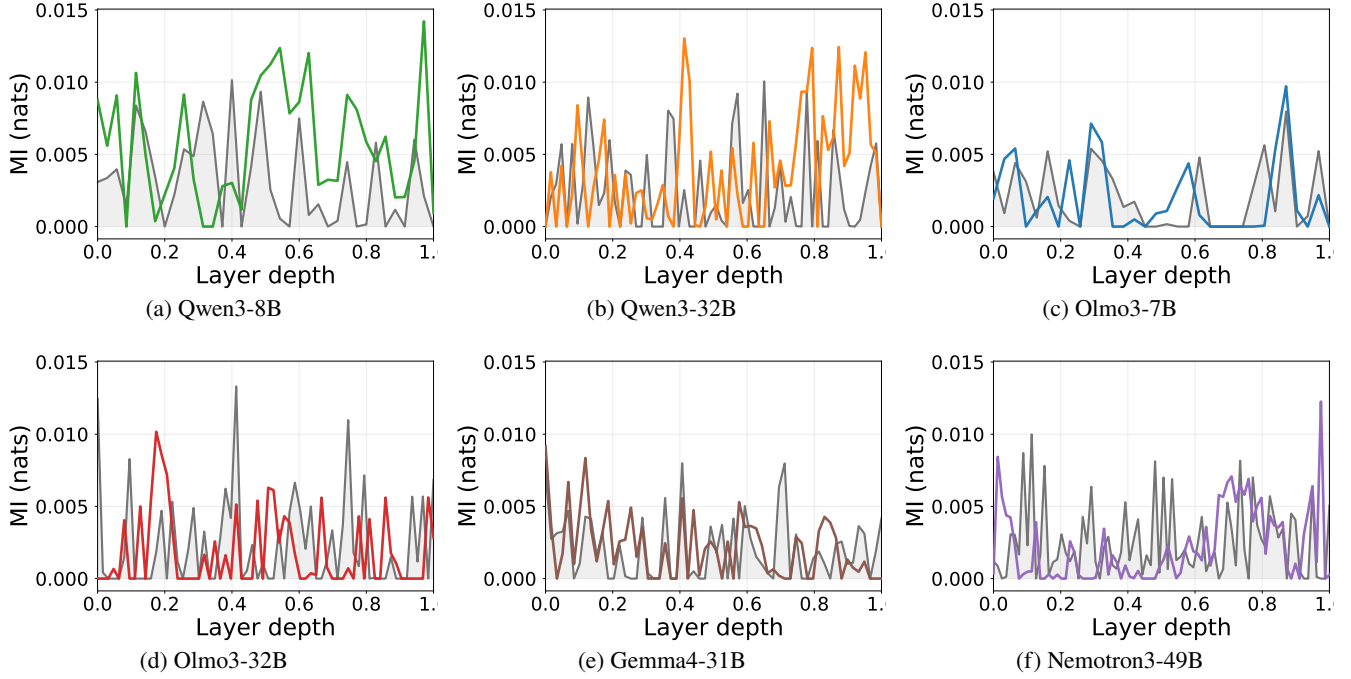


Figure 16: Mutual information between response-last probe score and judge score across the six models. Layer depth (normalized layer depth) denotes the layer index divided by the total number of hidden-state layers returned by the model.

Table 8: Peak per-prompt coupling between the linear probe $\hat{v}_\ell$ and the fine-grained judge J^{100} , taken over all layers, for the four reasoning checkpoints. Peak $|\rho_\ell|$ stays ≤ 0.17 for every probe readout, and peak MI stays ≤ 0.15 nats. The bottom block reports the same statistics for the `</think>`-token readout (Qwen only).

Readout	peak $ \rho_\ell $ (model / layer)	peak MI in nats (model / layer)
prompt-last	0.169 (Olmo3-32B/ L61)	0.149 (Olmo3-32B/ L12)
prompt-mean	0.080 (Olmo3-32B/ L63)	0.134 (Olmo3-32B/ L42)
response-last	0.135 (Qwen3-32B/ L8)	0.033 (Olmo3-32B/ L44)
response-mean	0.163 (Qwen3-32B/ L63)	0.034 (Olmo3-7B/ L21)
<code></think></code> , Qwen3-8B	0.072 (L6)	0.0081 (L34)
<code></think></code> , Qwen3-32B	0.087 (L62)	0.0129 (L61)

the ones placed further on the eval-like side of $\hat{v}_\ell$ should be the ones the model verbalizes on. Even with most judge scores at zero, this would yield clearly nonzero $|\rho_\ell|$ and MI. The values we observe are not consistent with that prediction, so the two measurements track different signals. The causal contribution of the probe direction to verbalization is the question addressed by the steering analysis in Section 4.

H Additional Steering Results

This appendix contains extra material for the steering analyses of Section 4. Appendix H.1 explains how the steering direction and its scale are computed. Appendix H.2 lists the layer and α ranges swept for each model. Appendix H.3 gives the unsteered baseline judge-score distributions used to compute $\Delta J(\alpha, \ell)$. Appendix H.4 gives the values of $|\alpha|$ at which outputs start to degrade. Appendix H.5 gives the best-layer statistics for each steering strength.

H.1 Direction Extraction and Scaling Details

Our steering pipeline uses asymmetric token aggregation across stages. In direction extraction, we compute contrastive directions from the final non-padding token representation (i.e., the decision-position hidden state, identified via the attention mask). In scaling, however, we estimate m_ℓ as the mean activation norm over all sequence positions in the calibration forward pass, without masking out padded tokens. Therefore, direction and magnitude are not matched to the same token subset; the scale can depend

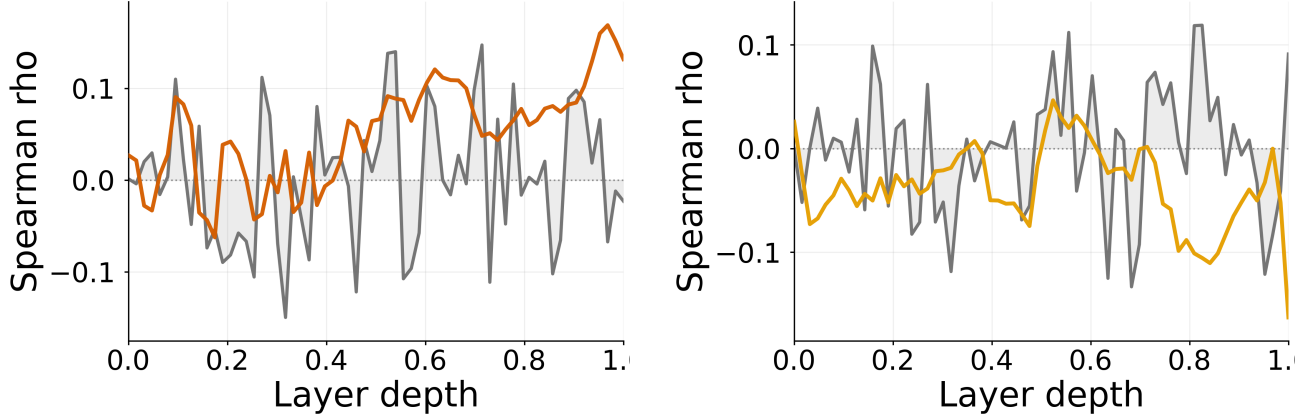


Figure 17: Layerwise Spearman ρ_ℓ between the linear probe $\hat{v}_\ell$ and the fine-grained judge J^{100} , on the two readouts that attain the highest peak $|\rho_\ell|$ in Table 8. **Left:** Olmo3-32B Think, prompt-last readout. The learned probe (coloured) tracks the random direction (grey, shaded to zero) closely; peak $|\rho_\ell| \approx 0.17$. **Right:** Qwen3-32B, response-mean readout; peak $|\rho_\ell| \approx 0.16$. Per-model plots for all four reasoning checkpoints, both metrics, and all four readouts are included in the supplementary material. Normalized layer depth denotes the layer index divided by the total number of hidden-state layers returned by the model.

Model(s)	Layer sweep	α sweep
Olmo3-7B Base	0–31	$\{-1.0, -0.8, -0.6, -0.4, -0.2, 0.2, 0.4, 0.6, 0.8, 1.0\}$
Olmo3-32B Base	22–41	$\{-1.0, -0.8, -0.6, -0.4, -0.2, 0.2, 0.4, 0.6, 0.8, 1.0\}$
Olmo3-32B SFT	22–41	$\{-1.0, -0.8, -0.6, -0.4, -0.2, 0.2, 0.4, 0.6, 0.8, 1.0\}$
Olmo3-32B DPO	22–41	$\{-1.0, -0.8, -0.6, -0.4, -0.2, 0.2, 0.4, 0.6, 0.8, 1.0\}$
Olmo3-32B Think	21–42	$\{-1.0, -0.8, -0.6, -0.4, -0.2, 0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$
Olmo3-7B Think	0–31	$\{-2.4, -2.2, \dots, 2.2, 2.4\}$ (step 0.2)
Olmo3-7B SFT	0–31	$\{-1.0, -0.8, -0.6, -0.4, -0.2, 0.2, 0.4, 0.6, 0.8, 1.0\}$
Olmo3-7B DPO	0–31	$\{-1.0, -0.8, -0.6, -0.4, -0.2, 0.2, 0.4, 0.6, 0.8, 1.0\}$
Qwen3-8B	0–35	$\{-0.8, -0.6, -0.4, -0.2, 0.0, 0.2, 0.4, 0.6, 0.8\}$
Qwen3-32B	0–63	$\{-0.6, -0.4, -0.2, 0.0, 0.2, 0.4, 0.6\}$

Table 9: Steering sweep configuration by model.

on padding/length statistics of the calibration batches. We report this implementation detail for transparency and treat it as a limitation of the current setup.

H.2 Sweep Details

Steering sweep details are shown in Table 9. For response judging, we evaluated a subsample of steered layers from the same sweeps: across the four judging runs (`every_2` for 32B models and `every_4` for 7B/8B models), we judged **113/310 model-layer combinations (36.45%)** in total.

H.3 Unsteered Baselines

Tables 10 and 11 report the unsteered baseline judge-score distributions used to compute the steering effect $\Delta J(\alpha, \ell)$ in Section 4. Table 10 covers the four final-stage models on which we run the main steering analysis (Qwen3-8B, Qwen3-32B, Olmo3-7B Think, Olmo3-32B Think); Table 11 covers all four Olmo training stages (Base, SFT, DPO, Think) for both 7B and 32B. For each model and stage, the baseline is the empirical judge-score distribution of natural completions on the same 1000 MASK prompts used for steering, falling back to $\alpha = 0$ when explicit unsteered rows are unavailable. Baseline averages $J(0, \ell)$ are close to zero across all settings, which contributes to the floor effect we discuss in Section 4.

H.4 Steering Breakpoints

For each model we sweep $\alpha \in \{-1.0, -0.8, \dots, 0.8, 1.0\}$ at every layer on a small number of samples. At sufficiently large $|\alpha|$, intervention can degrade model outputs to the point that the response becomes empty, repetitive, or non-language. Table 12 reports, for each model, the smallest $|\alpha|$ at which we observe any empty or degenerate output across the full layer sweep, and the fraction of (α, ℓ) cells affected at that magnitude. Only Qwen3-32B exhibits degeneration in the swept range.

Table 10: Unsteered baseline judge-score distribution for Qwen3 and Olmo3 Think models. N is the number of non-empty responses.

Family	Stage	N	Score 0 (%)	Score 1 (%)	Score 2 (%)	Score 3 (%)	Avg	Std
Qwen3-8B		1000	97.3	2.6	0.0	0.1	0.029	0.185
Qwen3-32B		1000	96.0	3.9	0.0	0.1	0.042	0.215
Olmo3 7B	Think	1000	93.5	6.3	0.0	0.2	0.069	0.276
Olmo3-32B	Think	1000	94.6	5.2	0.0	0.2	0.058	0.258

Table 11: Unsteered baseline judge-score distribution across Olmo3 7B/32B stages (falling back to $\alpha = 0$ when explicit unsteered rows are unavailable). N is the number of non-empty responses.

Family	Stage	N	Score 0 (%)	Score 1 (%)	Score 2 (%)	Score 3 (%)	Avg	Std
Olmo3 7B	Base	952	99.7	0.3	0.0	0.0	0.003	0.056
Olmo3 7B	SFT	1000	95.8	4.1	0.0	0.1	0.044	0.219
Olmo3 7B	DPO	1000	95.9	3.8	0.0	0.3	0.047	0.251
Olmo3 7B	Think	1000	93.5	6.3	0.0	0.2	0.069	0.276
Olmo3-32B	Base	962	99.1	0.9	0.0	0.0	0.009	0.096
Olmo3-32B	SFT	1000	96.2	3.7	0.0	0.1	0.040	0.211
Olmo3-32B	DPO	1000	95.9	4.0	0.0	0.1	0.043	0.217
Olmo3-32B	Think	1000	94.6	5.2	0.0	0.2	0.058	0.258

Table 12: Steering breakpoint summary. “First $|\alpha|$ with breakdown” is the smallest swept magnitude at which at least one (α, ℓ) response produces empty or degenerate output. “Cells affected at $|\alpha| = 1$ ” is the fraction of the (α, ℓ) grid affected at $|\alpha| = 1.0$.

Model	First $ \alpha $ with breakdown	Cells affected at $ \alpha = 1$	Notes
Olmo3-7B Think	–	–	No degeneration observed in sweep
Olmo3-32B Think	–	–	No degeneration observed in sweep
Qwen3-8B	–	–	No degeneration observed in sweep
Qwen3-32B	0.8	substantial	Empty outputs at certain (α, ℓ) combinations

H.5 Best Layer Statistics

We denote the mean judge score at steering strength α and layer ℓ by $J(\alpha, \ell)$. For each α , we select the best layer by maximizing $J(\alpha, \ell) + \sigma_J(\alpha, \ell)$, where $\sigma_J(\alpha, \ell)$ is the standard deviation of judge scores. We also report the distribution of non-zero judge scores (1, 2, 3) and the number of scored outputs used to compute these statistics. The results are shown in Figures 18–27.

Legend notes: In the average-score and scored-output panels, *All judged* means all responses that received a valid score from the judge. *No hit-max* removes responses that reached the generation length limit (`hit_max=true`); these outputs may be cut off, so their judged content can be incomplete. *No hit-max + Non-empty CoT* further restricts to responses with non-empty chain-of-thought (CoT) text. An $(n=0)$ marker indicates that no responses met that filter at that steering strength. In the average-score panel, this means the mean score is undefined at that point; in the scored-output panel, it means the count is exactly zero. The MASK dataset contains 1000 samples, so the maximum possible number of judged outputs at any point is 1000.

I Olmo Checkpoint Supplementary Material

This appendix provides supplementary analyses for the Olmo checkpoint experiments. Appendix I.1 reports the cosine similarity of the probe direction across stages. Appendix I.2 reports per-prompt coupling statistics across stages for both 7B and 32B. Appendix I.3 repeats the judge comparison of Appendix D across stages.